

OpenPantry — Why a Poisson GLM

Demand forecasting reference • Order Report • reports/order_report/forecast.php

THE SHORT ANSWER

Safety stock must be sized for the level being forecast

Safety stock is $Z \cdot \sqrt{\text{Var}}$. Sizing it requires the variance of the demand actually being ordered for — which, for a seasonal or growing item, is a level that has not recently been observed. A Poisson GLM supplies a variance that follows the forecast to that level. A constant-variance model cannot.

IN ONE LINE

Poisson lets the buffer breathe with the season. Gaussian freezes it at the yearly average.

THE PAYOFF

What level-responsiveness is worth

Bottled water moving 5/day in winter and 20/day in summer. Fourteen-day order, $Z = 1.65$, $\phi = 2$.

Season	Forecast	Poisson buffer	Constant-variance buffer
Winter — 5/day	70	19.5	30.9
Summer — 20/day	280	39.1	30.9

The Poisson buffer doubles as demand quadruples — spread grows as $\sqrt{4} = 2$. The constant-variance buffer was fitted once across the whole year and lands near the annual average in both seasons: roughly **58% too much stock in winter** and **21% too little in summer**, with the shortfall falling in the season most likely to run out.

THE THREE INGREDIENTS

Family, link, and design matrix are independent choices

Ingredient	Choice here	What it governs
family	quasi-Poisson	How observations scatter around μ . Sets the IRLS weight $w = \mu$ and $\text{Var} = \phi \cdot \mu$.
link	log	Maps the linear predictor η to μ . Makes effects multiplicative and keeps $\lambda > 0$ under extrapolation.
design matrix	[1, trend, sin/cos ...]	What η is composed of. Fourier seasonality lives here, gated on history span.

They compose, but none depends on the others — any one can be swapped. In the fit loop they are adjacent lines: $\mu = \exp(\eta)$ is the link; $w = \mu$ is the family. Log is the *canonical* link for Poisson, which is why the general IRLS weight collapses to exactly μ and the solver stays short enough to hand-roll in dependency-free PHP.

WHY THIS FAMILY

Why variance should scale with the mean here

Choosing Poisson means choosing the exponent 1 in $\text{Var} = \phi \cdot \mu^p$. That is not an arbitrary pick — it follows from the structure of how a pantry is actually used. Households arrive independently of one another, and each takes a variable number of items.

```
N ~ Poisson( $\lambda$ ) households arriving in the period
X_i items taken by household i (mean m, variance v)
T =  $\sum X_i$  total demand
E[T] =  $\lambda m$ 
Var[T] =  $\lambda(v + m^2)$ 
Var[T] / E[T] =  $(v + m^2) / m \leftarrow$  constant, independent of  $\lambda$ 
```

Variance comes out **exactly proportional to the mean**, and the constant of proportionality is ϕ . So ϕ is not a fudge factor bolted on to rescue a bad assumption — it is the average basket size inflated by its own variability. A pantry whose households take three items on average ($m = 3$, $v = 2$) predicts $\phi = 3.67$; $\phi = 1$ arises only in the degenerate case where every household takes exactly one item. Over-dispersion is the expected state, not a defect.

STATUS OF THIS CLAIM

This is a **structural argument, not an empirical finding**. It rests on assumptions — arrivals independent, basket size uncorrelated with the arrival rate — that are plausible for a pantry but have not been checked against the data.

The exponent is directly testable: bin the fitted buckets by μ , take the empirical variance within each bin, and regress $\log(\text{Var})$ on $\log(\mu)$. The slope is p — near 1 vindicates Poisson, near 2 points to Gamma or Tweedie, near 0 to Gaussian. That test needs fits spanning a range of demand levels, so it cannot run until well after the first fits appear.

Where the argument is thinnest: produce weighed in lbs. If the amount distributed is set by what the delivery truck brings rather than by what households request, that is a supply constraint rather than an arrival process, and there is no reason to expect an exponent of 1. Produce is the first place to look if that test ever comes back far from $p = 1$.

THE COMMON OBJECTION

Why ϕ does not make the Poisson redundant

ϕ is one scalar per item, fitted from Pearson residuals. It can rescale a relationship; it cannot create one. Posed the question that actually matters, a single number has no answer:

“ σ was 3.1 when demand ran 4/day. What is σ at 15/day?”

$\text{Var} = \phi \cdot \mu$ answers it in one multiplication:

```
from history:  $\phi = 3.1^2 / 4 = 2.4$ 
at 15/day: Var =  $2.4 \times 15 = 36$ 
 $\sigma = \sqrt{36} = 6.0$ 
```

	Supplies	Level-dependent?
Poisson	Variance proportional to μ	Yes — this is the part that transfers
ϕ	The constant of proportionality	No — one number, fitted from all buckets

So ϕ is not correcting a mistake in the Poisson; it fills the one blank the Poisson deliberately leaves. Pure Poisson hardcodes that constant at 1.0, which is usually wrong. Quasi-Poisson keeps the proportionality — the useful half — and measures the constant instead of assuming it.

Measured against known variability, the shape-plus-scale split reproduced true spread within **0.88–1.01×** across every demand pattern tested, including produce at $\phi \approx 0.06$ — a case where pure Poisson would have over-buffered by 4×

SCOPE

What the Poisson is *not* responsible for

RECURRING CONFUSIONS

λ staying positive is the log link, not the family. A Poisson GLM with an identity link can produce a negative fitted mean; a Gaussian GLM with a log link cannot. Multiplicative trend and seasonality are likewise the link.

The size of the variance is ϕ , not the family.

No Poisson probability mass function is ever evaluated. Quasi-likelihood specifies only the mean–variance relation, so quasi-Poisson is not, strictly speaking, a distribution at all.

The family's entire contribution is **$\text{Var}(\mu) = \mu$** — variance rises with level. Non-negativity and discreteness of the response are why that relationship tends to hold empirically for count data; they are not separate benefits the family delivers.

It follows that the Poisson earns nothing in a model that does not extrapolate. With no trend and no seasonality, μ never moves, and a variance that tracks μ is indistinguishable from a plain empirical one.

ALTERNATIVES

Other families that would also work

Family	$\text{Var}(\mu)$	Verdict
Gaussian	ϕ	Rejected — no μ in the formula, so the buffer cannot follow the forecast.
Gamma	$\phi\mu^2$	Rejected — requires $y > 0$; pantry data is full of zero weeks.
Negative binomial	$\mu + \mu^2/k$	Viable — handles zeros, but costs a second parameter, k .
Tweedie ($1 < p < 2$)	$\phi\mu^p$	Viable — best fit for produce in lbs, but p must be estimated.
quasi-Poisson	$\phi\mu$	Chosen — one scalar, closed-form from Pearson residuals.

Poisson also has an aggregation property the others lack. Because $\text{Var} = \phi \cdot \mu$ is linear in μ , variance over a multi-day horizon is $\phi \cdot \Sigma \lambda = \phi \cdot \text{Forecast}$ — a single multiplication. Under Gamma or Tweedie the horizon aggregation needs every day's rate individually.

Where this lives, and when it applies

```
Par Level = Forecast(LT) + Safety Stock
Forecast(LT) = Σ λ(today+d) for d = 1..LT
Safety Stock = Z × √( φ × Forecast(LT) )
```

Implemented in `reports/order_report/forecast.php` — `op_glm_fit_series()` fits the model, `op_project_forecast()` projects it over the lead time. Items without enough history fall back to a trailing average in `report_lib.php`, whose σ is measured over distribution days only so that closed-day zeros are not mistaken for demand volatility.

The model is gated on how much history each item has. Seasonal richness scales with span, so the capability arrives in stages:

Milestone	Date	Capability
First fit possible	2026-09-15	Intercept + trend only (harmonics = 0)
1 seasonal harmonic	2027-06-02	Annual cycle enters the model
2 seasonal harmonics	2027-12-29	Sharper seasonal shape

CURRENT STATUS

Scan history begins 2026-07-07. Until the first gate clears, every item runs the trailing-average fallback and no ϕ is computed anywhere — the S and G columns are hardcoded to 1.0. This is the gates working correctly on a young dataset, not a fault. The Poisson machinery contributes nothing until the model begins projecting levels it has not observed, which is when seasonality switches on.

Gates: span ≥ 60 days and ≥ 10 complete weekly buckets. Harmonics: 330 days for one, 540 for two. Training window capped at 1100 days. Dates derive from the earliest scan on record.